

Spectral Partitioning as a Deterministic Redistricting Construction Baseline

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Abstract

This paper studies spectral-style graph partitioning as a deterministic construction baseline for congressional redistricting. The paper focuses on the staged T.14 implementation: smoothed graph-coordinate approximation, population-balanced sweep cuts, recursive proportional target handling, and RPLAN audit sidecars through the BISECT CLI. The central claim is modest: spectral partitioning is a transparent deterministic baseline for comparison. RPLAN/RCTX sidecars verify declared profile and lineage, not fairness, legal compliance, exact optimality, or a replacement for METIS-family construction.

1 Introduction

Spectral graph partitioning is a natural deterministic baseline for redistricting because it exposes a continuous relaxation of the cut problem. This paper positions T.14 as a transparent construction method whose value is explainability, reproducibility, and comparison against stronger construction and optimization families.

The phrase “spectral” is used carefully here. The implemented T.14 slice is a deterministic spectral-style bisection baseline: it builds a graph coordinate by smoothing an anchored vector over adjacency, sorts units by that coordinate, and then selects a population-feasible sweep cut. It is not yet a full Lanczos or production eigensolver implementation of the exact Fiedler vector. That distinction is part of the contribution boundary rather than an implementation detail, because it tells the reader which claims are supported today: replayable construction, fixture behavior, and audit lineage, not spectral optimality.

2 Method

The staged method builds a deterministic spectral-style coordinate, sorts units by that coordinate, and evaluates population-balanced sweep cuts. Recursive calls pass proportional target fractions so odd and non-power-of-two district counts are handled without silently changing the intended seat allocation.

In the atlas framing, the candidates are not arbitrary graph cuts. They are prefix cuts induced by the Fiedler ordering. The ordering explains why the candidate set exists; the

sweep table explains why a particular prefix is selected or skipped. The downstream consequence of the selected root cut is a pair of recursive workloads with explicit left/right seat counts.

2.1 Reproducible Procedure

The constructor operates on the tract adjacency graph supplied by the BISECT state pipeline. Vertices carry population weights, edges encode adjacency, and the current implementation treats edge weights as the graph weights available from that pipeline. For each recursive subproblem:

1. Build the weighted graph induced by the current unit set.
2. Build a centered anchor vector from stable unit order and smooth it over adjacency for the configured iteration budget.
3. Normalize the coordinate after each smoothing pass and sort units by coordinate, breaking ties by stable unit identifier.
4. Sweep prefixes in that order, reject cuts outside the declared population tolerance, and select the feasible cut with minimum edge cut, breaking remaining ties by sweep index.
5. Recurse on the two induced subgraphs with the requested left and right seat counts until each leaf represents one district.

This procedure makes determinism an implementation contract: the same graph, weights, target fractions, and tie-breaking identifiers must produce the same assignment and lineage. The graph assumptions are deliberately narrow. Water gaps, islands, disconnected components, and alternative adjacency definitions belong to the input graph construction layer, and near-symmetric graphs can make the spectral coordinate weakly informative even when the run remains deterministic.

2.2 Odd-Seat Split Sketch

For non-power-of-two district counts, the important visual is the target tree, not a ratio label. A seven-district run first asks the root cut to create workloads for three and four seats. The four-seat side can then split evenly, while the three-seat side carries one odd split. The constructor therefore chooses each local bisection with the population target implied by the child seat counts, not by a universal 50/50 rule.

3 Implementation

The implementation lives in `bisect-apportion::spectral` and is exposed through `bisect state --structure spectral` and YAML `structure: spectral`. Final plan paths use the shared RPLAN/RCTX audit sidecar flow when emitted by the CLI.

Depth	Node target	Requested child targets	Local target fraction
0	7 seats	3 seats / 4 seats	3/7 then 4/7
1	3 seats	1 seat / 2 seats	1/3 then 2/3
1	4 seats	2 seats / 2 seats	1/2 then 1/2
2	2 seats	1 seat / 1 seat	1/2 then 1/2

Table 1: Recursive target propagation for $k = 7$. Each row describes the population target used by a local spectral sweep, so odd-seat recursion remains auditable rather than implicit.

The bisection kernel records the method name, sweep identity, iteration count, convergence flag, edge cut, population deviation, tolerance, target fraction, and parameter hash in `SpectralSummary`. The CLI recursion layer owns the split tree and passes the proportional target fraction for each recursive node. This crate boundary matters for review: `bisect-apportion` proves the deterministic local split contract on fixtures, while the CLI/RPLAN layer proves that recursive construction metadata is exported as a replayable package.

4 Audit Artifacts

When spectral construction is exported as a final plan, the paper follows the shared RPLAN and RCTX fixed point: assignments, algorithm lineage, parameter hashes, and validation results are written as sidecars that can be checked by the verifier without re-running the constructor. The audit claim is limited to the declared profile and does not imply political fairness or legal compliance. The audit package also does not prove that the spectral cut is unique, globally optimal, or preferable to another valid construction family.

For a reader inspecting an exported package, the fixed point is the pair “declared method identity plus replayable sidecars.” The certificate can say that the plan was emitted under the spectral profile with the recorded parameters and validation checks. It cannot say that the graph coordinate was the mathematically exact Fiedler vector, that the districting outcome satisfies state-specific legal standards, or that another construction family would not produce a better plan under some external objective.

5 Evaluation Plan

The first evidence layer is fixture correctness: path graphs, two-clique graphs, determinism, and recursive split handling. The current package layer adds a method-produced six-unit spectral fixture and a 100-unit synthetic benchmark package that exercise lineage, manifest hashes, and verifier acceptance. The next evidence layer should compare spectral construction against METIS, GeoSection, and clustering on selected real-data smoke cases.

The evaluation reading rule is therefore simple: fixture and package rows support construction correctness and audit replay, while comparison rows are research agenda items. The paper should not be read as evidence that spectral construction beats METIS, GeoSection, capacity clustering, or regionalization on plan quality; it creates the deterministic

Claim	Evidence now	Missing evidence	Status
Deterministic local split output	L0 path/two-clique/determinism tests	None for fixtures	Current
Odd- k recursion handling	Recursive target-fraction hardening	Broader k sweep	Current
RPLAN audit lineage	L1 sidecar checks, method package, and 100-unit benchmark smoke	Real-data export tran-script	Current
Quality vs. METIS/GeoSection	None in this paper	Smoke commands and outputs	Future
Runtime scaling	None in this paper	Benchmark script and hardware	Future

Table 2: Evidence ladder for T.14 claims. Fixture correctness, audit checks, and empirical plan quality are separate evidence levels.

Constructor	Primary role	Exposed failure mode	Audit surface
Spectral	deterministic baseline	weak cut or imbalance	lineage and validation
METIS	multilevel baseline	dependency and heuristic variance	manifest lineage
GeoSection	geographic baseline	geometric oversimplification	assignment checks
Capacity clustering	capacity baseline	infeasible capacity/repair	status sidecars
Regionalization	merge baseline	merge-order sensitivity	merge witnesses

Table 3: Planned constructor comparison framing. Quantitative entries remain future work until commands, data, and outputs are recorded.

baseline needed to run that comparison cleanly.

6 Limitations

Spectral partitioning is a relaxation. A valid and deterministic spectral plan does not imply global optimality, political fairness, or legal compliance beyond the declared audit profile.

The current implementation is also a staged spectral-style baseline rather than a full eigensolver. Its smoothing coordinate is useful because it is cheap, stable, and auditable, but claims that depend on exact Laplacian eigenvectors remain future work.

The method is also sensitive to graph construction choices. Disconnected components, weakly connected geography, nearly symmetric graphs, and edge weight conventions can change the spectral coordinate and therefore the sweep order. Those are inspectable inputs to the audit trail, but they are not solved by determinism alone.

7 Conclusion

T.14 adds a cheap, deterministic construction baseline to the BISECT family. Its publication value is clearest when it is presented as an auditable point of comparison rather than as the strongest available optimizer. The fixed-point audit contract makes the output reproducible under a declared profile, while the evidence ladder keeps fixture correctness, sidecar validation, and future empirical quality claims separate.

The next scientific step is comparative rather than rhetorical: run the same states, tolerances, and audit profiles across spectral, METIS, GeoSection, capacity clustering, regionalization, and flow construction, then publish only the differences supported by recorded commands and packages.

References

BISECT Project. T.14 spectral partitioning implementation spec, 2026. BISECT internal specification.